

EVAN VIENNEAU

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University of Alberta, Edmonton, Alberta, Canada

RESEARCH INTERESTS

Dark matter, indirect detection, quantum gravity, black holes

EDUCATION

Doctor of Philosophy (PhD) Physics

University of Alberta

September 2021 - Present

Edmonton, AB

- Using cosmological simulations to study the distribution and dynamics of dark matter halos in the context of direct and indirect detection, for a variety of astrophysical targets and dark matter models
- Using numerical methods to derive characteristics of quantum-corrected black holes modeling singularity resolution and dynamical horizons

Honours Bachelor of Physics Co-op

McMaster University

September 2016 - April 2021

Hamilton, ON

- Minor in computer science
- Graduated with an A-
- Senior thesis entitled "Properties of Low Mass Disc-Dominated Galaxies in Voids, Filaments, and Groups"

PUBLICATIONS

Dark matter decay signals in cosmic filaments

arXiv:2504.08025v1, to be submitted to JCAP

April 2025

A generalized uncertainty-inspired quantum black hole

Published in: Phys.Rev.D 111, 024048 (2025)

January 2025

Significant impact of Galactic dark matter particles on annihilation signals from Sagittarius analogues

Published in: JCAP 10 (2024) 019

October 2024

The local dark matter distribution in self-interacting dark matter halos

Published in: JCAP 02 (2023) 040

February 2023

VERTICO: The Virgo Environment Traced In CO Survey

Published in: Astrophys.J.Suppl. 257 (2021) 2, 21

November 2021

PRESENTATIONS

2024 Canadian Association of Physicists Congress

Western University

May 2024

London, ON

- First place student talk in particle physics division
- Presented preliminary results regarding the impact of the Large Magellanic Cloud on dark matter indirect detection signals

Loops'24 International Conference on Quantum Gravity

Florida Atlantic University

May 2024

Fort Lauderdale, FL

- Presented our paper, 'A generalized uncertainty-inspired quantum black hole'

2023 Canadian Association of Physicists Congress*University of New Brunswick*

June 2023

Fredricton, NB

- Presented our paper, 'The local dark matter distribution in self-interacting dark matter halos'

VERTICO Collaboration Meeting*McMaster University*

December 2020

Hamilton, ON

- Presented my research on the radio continuum of Virgo cluster galaxies undergoing interactions and/or ram pressure stripping

Department of Physics and Astronomy Poster Fair*McMaster University*

May 2020

Hamilton, ON

- Presented my research on the environmental dependence of galaxy morphology via a talk and poster

SNO+ Bi-Yearly Collaboration Meeting*SNOLAB*

March 2019

Lively, ON

- Presented my research on calibration source umbilical behaviour in scintillator fluid

SCIENTIFIC VISITS

2023 Canadian Astroparticle Physics Community Meeting*QueensU*

August 2023

Kingston, Ontario

- Attended a variety of presentations, focused on communicating the research progress achieved by the Canadian astroparticle physics community

2022 Tri-Institute Summer School on Elementary Particles*TRIUMF*

July 2022

Vancouver, British Columbia

- Attended a diverse array of lectures to learn, engage and network with leading experts in the field

WORK EXPERIENCE

Astrophysics Researcher and Beta Tester*McMaster University*

September 2020 - April 2021

Hamilton, ON

- Acted as beta-tester for the development of a state-of-the-art cloud-based computing environment named ARCADE
- Worked with an ALMA imaging pipeline to optimize the processing of interferometric radio continuum data from the VERTICO survey of galaxies
- Analysis of this data lead to potential evidence for enhanced radio continuum emission on the leading edge of infalling Virgo cluster galaxies

Physics Pedagogy Researcher*McMaster University*

July 2020 - August 2020

Hamilton, ON

- Developed web-based Python programming bootcamp for incoming science undergraduates
- Investigated ways to better incorporate computation into the undergraduate physics curriculum
- Supported instructors in the organizational and logistical side of running online courses

Neutrino Detection Researcher*SNOLAB*

January 2019 - June 2019

Lively, ON

- Ran quality assurance tests on scintillator samples using density, turbidity and UV absorption meters
- Monitored state of cover gas bags in detector cavity to maintain correct nitrogen levels
- Worked detector operator shifts monitoring and gathering detector data

- Performed independent research on the performance of calibration source umbilicals in scintillator fluid

OTHER RESPONSIBILITIES

Graduate Student Physics Association - Ambassador

University of Alberta

January 2024 - December 2024

Edmonton, AB

- Act as the main liaison between the graduate student physics association and the graduate student association

Graduate Student Physics Association - VP Finance

University of Alberta

January 2023 - December 2023

Edmonton, AB

- Oversee GPSA finances and manage GPSA bank account
- Act as signing authority
- Support fundraising activities

Physics Graduate Student Association - YUGSA representative

York University

September 2021 - December 2021

Toronto, ON

- Networking and community development

TEACHING ASSISTANTSHIP

PHYS144 - Newtonian Mechanics

University of Alberta

September 2024 - December 2024

Edmonton, AB

- Running and marking of lab sessions

PHYS146 - Fluids and Waves

University of Alberta

January 2024 - April 2024

Edmonton, AB

- Overseeing and facilitating semester long student research projects in a variety of physics subfields

ENPHYS131 - Mechanics

University of Alberta

January 2023 - April 2023

Edmonton, AB

- Running and marking of lab sessions

PHYS130 - Wave Motion, Optics and Sound

University of Alberta

September 2022 - December 2022

Edmonton, AB

- Running of large class tutorial sessions and marking

PHYS124 - Particles and Waves

University of Alberta

September 2022 - December 2022

Edmonton, AB

- Running of large class tutorial sessions and marking

PHYS1421 - Physics with Life Science Applications 1

York University

May 2022 - June 2022

Toronto, ON

- Invigilation of midterms and exams
- Marking of class assignments, midterms and exam

PHYS2030 - Computational Methods for Physicists and Engineers

York University

January 2022 - April 2022

Toronto, ON

- Marking of weekly assignments

PHYS1422 - Physics with Life Science Applications 2

York University

January 2022 - April 2022

Toronto, ON

- Marking of class tests

PHYS1411 - Physics Fundamentals 1

York University

September 2021 - December 2021

Toronto, ON

- Running of lab tutorial sessions and marking labs